A high-accuracy phase-based ranging solution with Bluetooth Low Energy (BLE)

Conference Paper · April 2019

DOI: 10.1109/WCNC.2019.8885791

Pouria Zand
Imec the Netherlands

Jac Romme
Imec the Netherlands

W.F. Frank Pasveer
ASML

Guido Dolmans
Imec the Netherlands

A high-accuracy phase-based ranging solution with Bluetooth Low Energy (BLE)

Pouria Zand, Jac Romme, Jochem Govers, Frank Pasveer and Guido Dolmans
IMEC, Eindhoven, The Netherlands
e-mail:{pouria.zand, jac.romme, jochem.govers, frank.pasveer and guido.dolmans}@imec.nl

***Abstract*—Nowadays, indoor positioning and asset tracking have become popular and essential for many applications and use-cases. As Bluetooth Low Energy (BLE) is widely deployed in smart tags, smart-phones, and smart-devices, adding functionality to support localization and asset tracking is crucial. In order to locate a device, either the angle or range to the device needs to be calculated. In this paper, we focus on a phase-based solution to calculate the range between devices. We introduce a multi-carrier phase-based ranging solution compatible with the BLE standard that utilizes BLE channel hopping to exchange tones in the entire 2.4 GHz frequency band to mitigate the multi-path fading problem. We recognize that in the BLE standard, slow channel-hopping, long packet size and frame spacing between two consecutive communications (named T_IFS) affect ranging error/accuracy, in the case of crystal offset and phase-noise. Therefore, we introduce a new mathematical model to analyze the impact of the BLE link layer protocol on the ranging error. Finally, we evaluate the accuracy of our proposed solution through modelling, by considering the results of real experiments and by validating the correctness of our mathematical model for the ranging error.**

***Index Terms*—Bluetooth Low Energy (BLE), ranging, localization, phase-measurement, channel hopping, CTE**

I. Introduction

Nowadays, many devices are connected to each-other and to the Internet, a set up known as the Internet-of-Things (IoT). In some IoT indoor applications, it is desired and essential to know the distances between these devices and to locate them. These devices use various wireless technologies to communicate with each other. Bluetooth Low Energy (BLE) is one of the most famous and popular ones; it can be found in many tags, smart phones, beacons, and smart devices. As such, the application of a BLE radio to perform ranging and to locate devices has increasingly become inevitable.

Ranging solutions can be classified into Received Signal Strength Indicator (RSSI)-based, time-based and phase-based approaches. In RSSI-based solutions, the receiver of the signal will calculate its distance to the transmitter based on the attenuation of the transmitted signal over a distance. RSSI-based solutions suffer from multi-path fading and many researchers use finger-printing [1] often in combination with a the Kalman filter [2] to mitigate this problem. In the time-based solution, the distance between two devices is calculated based on propagation delay. Time-based solutions can be classified into Time of Arrival (ToA) [3], Time Difference of Arrival (TDoA) [4] and the Round Trip Time of Flight (RT-ToF) [5]. However, time-based solutions require a large signal bandwidth in order to have acceptable accuracy in multipath environments, which is incompatible with BLE.

In the phase-based solution, the amount of signal-phase shifts between transmitter and receiver is used to calculate the distance between them [5]. To mitigate the multi-path problem, two devices measure phase changes over multi-frequencies[1] to calculate the distance between them. This procedure is entitled Multi-Carrier Phase Difference (MCPD). In this paper, we introduce a multi-carrier approach compatible with the BLE standard that can combat multi-path fading and does not require tight synchronization between devices.

To that end, we first introduce a new phase-based ranging solution compatible with BLE. Our ranging solution works in both connection-oriented and connectionless modes of operation. The BLE slow channel-hopping technique is used to perform multi-carrier tone-exchange in the entire 2.4 GHz frequency band (80 MHz). However, slow channel-hopping, as well as other BLE link-layer regulations such as frame spacing between consecutive packet exchange, can affect the accuracy of our proposed solution. Therefore, we also study the extent to which ranging accuracy is affected by link-layer regulations in the presence of phase-noise and crystal offset. To that end, we introduce a mathematical model that analyzes the impact of crystal offset on ranging accuracy. Finally, we perform experiments and study mathematical model outcomes to confirm the results of our mathematical model and to examine the effect of BLE standard regulations on the accuracy of the proposed ranging solutions.

Altogether, we aim to:

- Introduce a new multi-carrier phase-based ranging solution that is compatible with BLE and supports both connection-oriented and connectionless modes of operation.
- Introduce a new mathematical model to analyze the effect of BLE's slow channel-hopping and phase-sampling time on ranging error, in the presence of crystal-offset.
- Evaluate the impact of phase-noise and crystal offset on ranging accuracy through mathematical modeling and hardware experiments.

II. Related works

In [6], [7] and [8] the authors introduce a solution to calculate the Angle of Arrival (AoA) and Angle of Departure

[1] In this article, the terms frequency, carrier, channel and tones are used interchangeably.

(AoD) using BLE, where new positioning information is added to BLE packets after Cyclic Redundancy Check (CRC) named *Tail bits* that are used to measure AoA/AoD. Recently, Bluetooth Core Specification 5.1 [9] specified AoA and AoD features for BLE. In the new specification, the positioning information, namely Constant Tone Extension (CTE), is a constantly modulated series of unwhitened 1s. In the AoA approach, the receiver samples phase and amplitude from its antenna array while receiving signals from the target device and switching between antennas. In the AoD approach, the direction finder device samples phase and amplitude while receiving the signal from the Beacon, which performs antenna switching during signal broadcasting. The direction finder device can estimate the AoD upon obtaining the information about antenna switching patterns from the transmitter. Unlike [6] and [8] that only need unidirectional communication, our proposed phase-based solution requires bi-directional, half-duplex communication. Section IV describes our proposed ranging solution in more detail.

In [10] and [11], the authors propose a phase-based ranging method for a fast frequency hopping system. The proposed solution is a kind of anti-interference ranging method that offers strong resistance to radio interference, by utilizing fast frequency hopping technology. Similar to our solution, [10] and [11] also use the information obtained by all the available channels to provide a high accuracy estimation. However, their solution is based on calculating the phase changes of multi-subcarriers (two subcarriers in [11] and m subcarriers in [10]) within each channel. Moreover, their method does not conform to the BLE standard, unlike our solution.

A phase-based ranging solution is proposed in [12], where multiple devices are involved in the ranging procedure based on Radio Interferometric Positioning System (RIPS). In RIPS, two transmitters send two pure sine waves at two close frequencies and two receivers measure the relative phase offset of the received carriers [13], [14]. Then, a third party can calculate a linear combination of the distances between these four nodes, and finally estimate their relative position. Unlike our proposed solution, in which only two nodes participate in ranging, RIPS requires four nodes. This makes this method a relatively complex one. In addition, our solution is based on bi-directional carrier exchange between two nodes, while RIPS is based on uni-directional carrier transmission.

In [15], the authors describe phase-based ranging of (multiple) RFID tags, where passive tags communicate with the reader by using back-scatter communication. However, this solution [15] does not conform with the BLE communication standard.

In [16], the authors propose a multi-carrier phase difference solution based on IEEE 802.15.1 and, preferably, IEEE 802.15.4 radio that works in a proprietary mode. The initiator and the reflector will first agree on a pseudo-random channel hopping sequence, as part of the initiation procedure. Then, in the phase-measurement stage, they will exchange carrier signals on these agreed channels. The proposed solution in [16] only exchanges un-modulated carrier signals and the mechanism does not conform with the packet structure of either IEEE 802.15.1 or IEEE 802.15.4 data link layer specification, including the channel hopping mechanism.

III. Principle of Distance Estimation using MCPD

It has been well-established that the phase-shift introduced by a pure Line-of-Sight (LOS) radio channel on a radio signal is a linear function of both frequency (f) and range (r), i.e.

$$\phi(f, r) = -2\pi f r / c_0 \pmod{2\pi} \tag{1}$$

where c_0 is the speed of light in vacuum. Hence, by measuring the slope of the phase as a function of frequency, the distance between the radio devices can be measured and used in the MCPD principle.

In MCPD-ranging two roles are defined, namely initiator and reflector. The initiator is a device that starts the ranging-procedure. The reflector is a device that responds to the initiator. In this work, we assume that the initiator has the BLE master role and the reflector has the BLE slave role, in connection-oriented mode. After a handshake, which is also used to realize coarse time synchronization, the initiator transmits its Local Oscillator (LO) signal, i.e. a constant tone[2], to the reflector on the first channel (f_0). The reflector performs phase (or I/Q) measurement on the received carrier, the result of which depends on both the reflector's and the initiator's LO and distance between them. Following this, the reflector sends back a constant tone, i.e. its LO, to the initiator on the same channel, so that the initiator can perform a phase measurement. This procedure is repeated on K_f channels, where we assume, for sake of simplicity, a uniform channel spacing denoted by Δ_f. The initiator and the reflector perform carrier transmission and reception over the entire frequency band (e.g., 80 MHz at 2.4 GHz) to mitigate multi-path fading, as is shown in Figure 1. Afterwards, the reflector will send the measurement results over the entire frequency band to the initiator, allowing the initiator to calculate the range [16].

The MCPD method only requires coarse synchronization between the clocks of the initiator and reflector. The phase measured at the reflector is relative to its own LO signal phase and is sent to the initiator, so that the initiator can use it as a correction factor [16].

Let us define Δ_t to be the time-offset between initiator I and reflector R, then, at R we will measure the following phase

$$\phi_R(f_k, r) = -2\pi f_k \left(\frac{r}{c_0} - \Delta_t \right) - \theta \pmod{2\pi} \tag{2}$$

and after changing roles,

$$\phi_I(f_k, r) = -2\pi f_k \left(\frac{r}{c_0} + \Delta_t \right) + \theta \pmod{2\pi} \tag{3}$$

[2] In this article, the LO signal referred to the continuous wave originates often directly from the local oscillator signal, however, as example, it can also be obtained by FSK-modulating with a constant input (all 1s or 0s), named CTE.

where θ is the phase difference between the LO signal of the initiator and the reflector. After summing the phase measured at I and R, the signal has traveled twice the distance r between I and R, i.e. the phase difference will be

$$\phi_{2W}(f_k, r) = \phi_R(f_k, r) + \phi_I(f_k, r) = -4\pi f_k r / c_0 \pmod{2\pi} \tag{4}$$

where (4) is independent of the time offset. The same will happen if we introduce a phase-offset between the LOs of I and R. However, this implies that the phase-measurements taken at I and R have to be available at one place, typically the initiator.

There is still a half-wavelength ambiguity w.r.t. r in (4), which is resolved by measuring the phase shift at two (or more) distinct tones. In this case, the range ambiguity will depend on the frequency difference of both tones (Δ_f),

$$\Delta_\phi = \phi_{2W}(f_1, r) - \phi_{2W}(f_0, r) = -\frac{4\pi\Delta_f}{c_0} r \pmod{2\pi} \tag{5}$$

which does not depend on absolute frequency.[3] Hence, the estimate for r becomes

$$\hat{r} = -\frac{c_0}{4\pi\Delta_f}\hat{\Delta}_\phi \ (\mathrm{mod}\ \frac{c_0}{2\Delta_f}) \tag{6}$$

where the range-ambiguity is equal to $c_0/(2\Delta_f)$. The procedure can be extended to K_f tones. Let $\Delta_\phi[k]$ be the phase difference between the k-th tone and a $k-1$-th tone, such that

$$\hat{\Delta}_\phi = \frac{1}{K_f - 1}\sum_{k=1}^{K_f - 1}\hat{\Delta}_\phi[k] \pmod{2\pi} \tag{7}$$

which is the same as the phase difference between the first and last tone *after phase-unwrapping*, i.e.

$$\hat{r} = \frac{c_0}{4\pi\Delta_f}\frac{\phi_{2W}(f_{K_f}, r) - \phi_{2W}(f_0, r)}{K_f - 1} \tag{8}$$

Using multiple tones, the bandwidth has effectively been increased by a factor $K_f - 1$, without reducing the unambiguous range. As a result, the MCPD range-estimate will be less sensitive to phase errors.

In the BLE standard, the maximum frequency separation between tones is around 80 MHz, which is divided into 40 channels with 2 MHz bandwidth. Three of these channels are dedicated to advertising, the remaining 37 to data. Thanks to the BLE's pseudo-random frequency hopping and MCPD techniques, and by exchanging 37 carriers over those narrow-band channels, the system can be seen as a wide-band solution. This helps to improve multi-path robustness, while keeping the 'instantaneous' bandwidth small. Furthermore, the system is designed for half-duplex devices and uses narrow-band constant-envelope signals, allowing for the usage of BLE-compliant radio chips. The 2 MHz channel bandwidth relates to a 75 meter range ambiguity, which is well above the typical coverage of BLE in terms of distance. The only atypical requirement, from a BLE-perspective, is that the LO generation needs to be active (and in lock) when switching from transmitter (TX) to receiver (RX) and vice-versa.

[3]So far the phase-shift induced by switching I from TX to RX and, conversely, for R has been neglected. Assuming that (nearly) similar phase shifts will take place for f_0 and f_1, (5) will (almost) be independent of TRX switching induced phase-shift due to the subtraction.

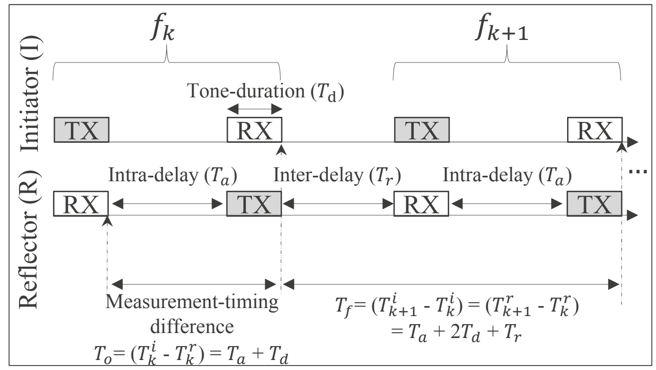


Fig. 1. Multi-carrier phase-based ranging principle

IV. A PROPOSED RANGING SOLUTION COMPATIBLE WITH THE BLE STANDARD

In this section, we describe our proposed solution to perform multi-carrier phase-based ranging based on the BLE standard.

A. Connection-oriented and connectionless approaches

The proposed multi-carrier ranging solution requires carrier exchange between the initiator (or anchor) and the reflector (or target device) in the entire 2.4GHz ISM frequency band. The solution works in both connection-oriented and connectionless modes. In the **connection-oriented** mode, the initiator and reflector establish a connection with one another. The periodic communication happens in the *connection event* window, while the subsequent connection event happens in a new channel. The two devices calculate the next channel, based on their pseudo-random frequency hopping sequence pattern: $f_{k+1} = (f_k + hop)\ mod\ 37$, where f_k is the frequency channel for current communication, f_{k+1} is the frequency channel for subsequent communication, hop is the hopping parameter ranging from 5 to 16 and 37 comprises all of the channels. The period of connection is determined by the *connection interval*. Its value ranges from 7.5 ms to 4 secs (with increments of 1.25 ms). Our proposed ranging solution utilizes the BLE channel hopping to exchange constant tones over the entire frequency band. In our proposed solution, two BLE commands are required (e.g., Ranging_IND) to carry CTE (i.e., tail bits) at the end of the packet after the Cyclic Redundancy Check (CRC). Figure 2 depicts this process by highlighting three cycles of master and slave (or initiator and reflector) communications. At the end of each cycle, both initiator and reflector hop to a new channel.

In the **connectionless** mode, unlike in connection-oriented mode, the initiator and reflector do not establish any connection. However, phase-based ranging requires the initiator and reflector to bi-directionally communicate with one another and

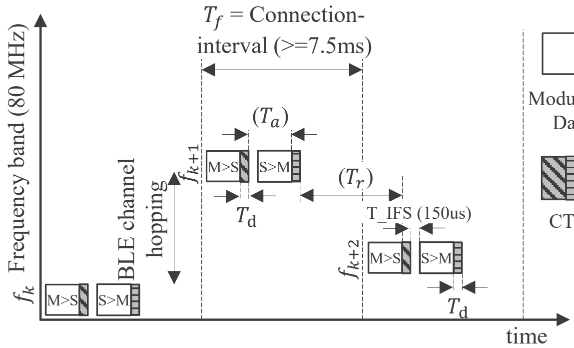


Fig. 2. Connection-oriented ranging solution, where BLE master (M) has the initiator role and BLE slave (S) has the reflector role

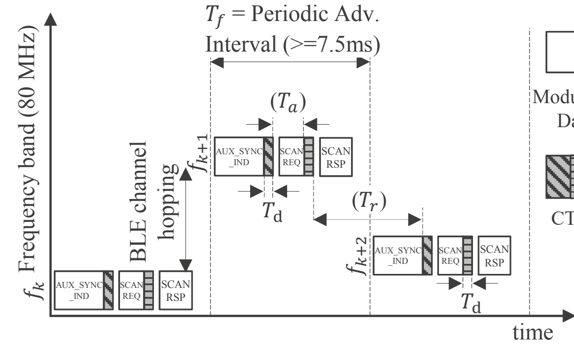


Fig. 3. Connectionless ranging solution, where BLE advertiser has the initiator role and BLE scanner has the reflector role

exchange tones. Fortunately, there are ways to establish bidirectional communication in connectionless mode. One example is having the scanner device send out a scan request and let the advertising device reply with a scan response. In addition, in Bluetooth 5, thanks to periodic advertisement, the advertiser periodically broadcasts (AUX_SYNC_IND) over the 37 data channels by performing channel hopping with an interval from 7.5 ms to 81.91875 s. We assumed the AUX_SYNC_IND can carry the CTE to enable ranging. Any scanning device that receives periodic advertisements can send a SCAN_REQ command that also carries the CTE and can perform ranging with the desired broadcasting device. However, AUX_SYNC_IND are not scannable and some modifications are required in the new BLE specification to allow other scanning devices to send SCAN_REQ toward the broadcaster. Figure 3 shows a sample of our BLE connectionless ranging solution, depicting three cycles of broadcaster and observer/scanner (or initiator and reflector) communication.

In the connection-oriented mode, the entire proposed multi-carrier ranging solution duration depends on the connection interval value. Exchanging the signals on the entire frequency band takes at least 277.5 ms. In order to locate a position of a target device based on the trilateration technique, at least three anchors have to perform ranging with a target device that takes at least 832.5 ms. For applications that need to perform real-time ranging with numerous devices or anchors, this solution may not be fast enough.

B. Challenges caused by intra- and inter-delay between carriers

In both connection-oriented and connectionless modes, the proposed BLE based ranging solution exchanges CTEs between reflector and initiator over the entire frequency band. In this paper, the delay between a transmitted and received tone by an anchor at channel f_k is called *intra-delay* (or T_a). In other words, *intra-delay* comprises the amount of time that is taken up when the reflector performs a phase measurement on the carrier sent by the initiator, and when the initiator subsequently does the same on the response carrier it received from the reflector. The time difference between the tone exchange in channel f_k and the tone exchange in the next channel f_{k+1} is called *inter-delay* (or T_r). In Figure 1, the intra-delay and inter-delay are shown. The intra-delay is equal to the sum of the length of the packet carrying the CTE and the Time Inter Frame Space (T_IFS), which is the time interval between consecutive packets on the same channel index. Conforming to the Bluetooth standard, the packet length varies from 44 to 2120 μs and T_IFS is equal to 150 μs. The inter-delay is determined by the connection-interval in the connection-oriented mode or by the periodic advertisement interval in the connectionless mode. Figure 2 shows the intra-delay and inter-delay in our BLE-based ranging solution.

C. A mathematical model to study the effects of intra- and inter-delay on ranging error

The MCPD approach is known to be relatively insensitive to frequency offset [17]. However, to comply with the time-frequency division scheme in Bluetooth low-energy (BLE), the time-offsets between sampling-instant/time will be much larger.

In this section, we derive a mathematical model to analyze the impact of crystal offset on the ranging accuracy/error. We will assume that a single crystal is used for both LO- and clock-generation, which is the case for most BLE-transceivers. As a result, the actual frequencies and timing will scale by a factor $(1+\eta)$ accordingly, where η is the crystal offset and is typically expressed in part-per-million (ppm). For example if the crystal offset is measured to be 1 ppm, it means that $\eta = 10^{-6}$. As both the initiator and reflector use different crystals, η_i, η_r will be the crystal offset at the initiator and reflector, respectively.

In case of crystal-offset, it can be shown from (2), (3), and (4) that

$$\begin{aligned}\phi_{2W}(f_k, r) = &-2\pi(f_k^i + f_k^r)\frac{r}{c_0} \\ &+ 2\pi(f_k^i - f_k^r)(T_k^r - T_k^i)\end{aligned} \tag{9}$$

where f_k^i and f_k^r are the actual frequencies generated by respectively the initiator and the reflector, and T_k^i, T_k^r represent the actual time at which the phase is measured at the initiator and reflector, respectively.

As stated before, the actual frequencies and time-offset will scale in the presence of crystal-offsets. The actual frequencies will scale accordingly $f_k^i = (1+\eta_i)f_k$ and $f_k^r = (1+\eta_r)(f_k+F)$, where f_k represents the desired frequency for the k-th channel with $f_k = f_c + k\Delta_f$ and f_c is the frequency carrier. For completeness, we added a variable F, which denotes any programmed frequency to model a software-controlled frequency offset. The measurement-timing is defined as follows; $T_k^r = (1+\eta_r)t_k$, $T_k^i = (1+\eta_i)(t_k+T_o)$ with $t_{k+1}-t_k = T_f$, where T_o is the time difference at which measurement is taken as the initiator and the reflector and T_f is the BLE connection/advertisement interval that ranges from 7.5 ms to 4 s (with increments of 1.25 ms).

After substituting these definitions into (9) in combination with (5), (6) and (7), the ranging error is

$$e_r = \hat{r} - r = \alpha_f T_f + \alpha_o T_o \tag{10}$$

with

$$\alpha_f = -\frac{c_0}{2}\left((\eta_r-\eta_i)^2\left(\frac{f_c}{\Delta_f}+K_f\right)+(1+\eta_r)(\eta_r-\eta_i)\frac{F}{\Delta_f}\right) \tag{11}$$

$$\alpha_o = -\frac{c_0}{2}(1+\eta_r)(\eta_r-\eta_i) \tag{12}$$

When using the definition shown in Figure 1, we can relate these values to the intra- and inter-delay. Figure 1 shows that $T_o = T_a + T_d$ and $T_f = T_a + T_r + 2T_d$, causing the ranging error to be

$$e_r = (\alpha_f+\alpha_o)T_a + (\alpha_f)T_r + (2\alpha_f+\alpha_o)T_d \tag{13}$$

Using some practical values for BLE (f_c=2.4GHz, Δ_f=2MHz, K_f=40), we can now quantify the impact of crystal-offset on the ranging error. Firstly, α_0 tends to be much larger than α_f. For instance, if we assume F=0 kHz and a crystal-offset of at most $\pm$20 ppm, we see that α_0 is at least a factor 20 larger than α_f. In this case, the η difference is at most $40*10^{-6}$, causing the quadratic term in (11) to be small compared to the other terms, despite the large scaling term ($f_c/\Delta_f + K_f$=1240). As a result, the range error is more sensitive to intra-delay (T_a) than to inter-delay (T_r). Even when the crystal offset is at its highest at 40 ppm, α_0 is still a factor 13 larger than α_f.

This means that for BLE systems, both (11) and (12) can be approximated, by ignoring the quadratic term in (11) and using the fact that $(1+\eta_r)\approx 1$, so that

$$\tilde{\alpha}_f = -\frac{c_0}{2}(\eta_r-\eta_i)\frac{F}{\Delta_f} \tag{14}$$

$$\tilde{\alpha}_o = -\frac{c_0}{2}(\eta_r-\eta_i) \tag{15}$$

The attentive reader may have noticed that we did not use the BLE pseudo-random hopping patterns, but undertook a linear sweep of all channels instead, where k-th channel is $f_k = k\Delta_f$. Using a pseudo-random pattern, this will equal $f_k = ((f_{k-1}+hop)\ mod\ 37)\Delta_f$. As all 37 channels are still in use, following a (off-line) re-ordering of the ϕ_{2W}'s, we can still apply the MCPD principle. The hopping does not affect the intra-delay, but the time-interval/inter-delay between tones that are Δ_f apart from one another can be significantly larger and may be negative. Overall, the effective inter-delay is still reasonably small and (very) similar to the earlier-mentioned results.

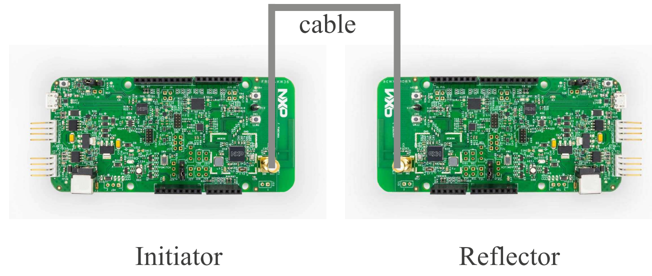


Fig. 4. Experiment setup for wired (controlled environment) scenario

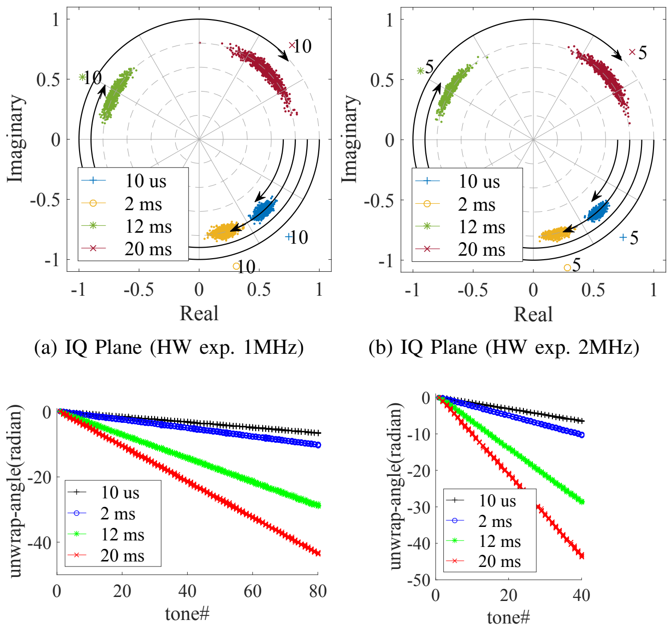
V. Comparison of Experimental and Analytic Results on Intra- and Inter-Delay

In this section, we evaluate the effect of intra- and inter-delay on ranging accuracy using a hardware experiment and compare it with the expectation using the analytical equations of Sec. IV-C. Figure 4 illustrates the experiment setup, during which two FRDM-KW36 development kits with an integrated 2.4 GHz transceiver that supports BLE v5 with a 32 MHz reference oscillator crystal were used. The crystal offset of the initiator and reflector has been measured to be 7.25 and 4 ppm, respectively. Furthermore, BLE has a frequency step (Δ_f) or channel bandwidth of 2 MHz. However, in the experiment and in the analytical equations, we study two frequency steps (i.e., 1 MHz and 2 MHz) to be able to validate our analytical equations and relation to Δ_f more precisely.

A. Intra-delay (T_a) increase and the effect of crystal offset on ranging error

In this part, we evaluate the effects of increased T_a on ranging errors due to crystal offset. We collected 1000 measurement values for each of the 80 tones with a frequency step of 1 MHz. For clarity, without affecting the ranging results, we rotated the IQ-result of all tones to ensure the first tone is on the positive x-axis. For better illustration, Figure 5(a) only depicts the result for tone #10. However, it does so for four different T_a-values, namely 10μs, 2ms, 12ms and 20ms.

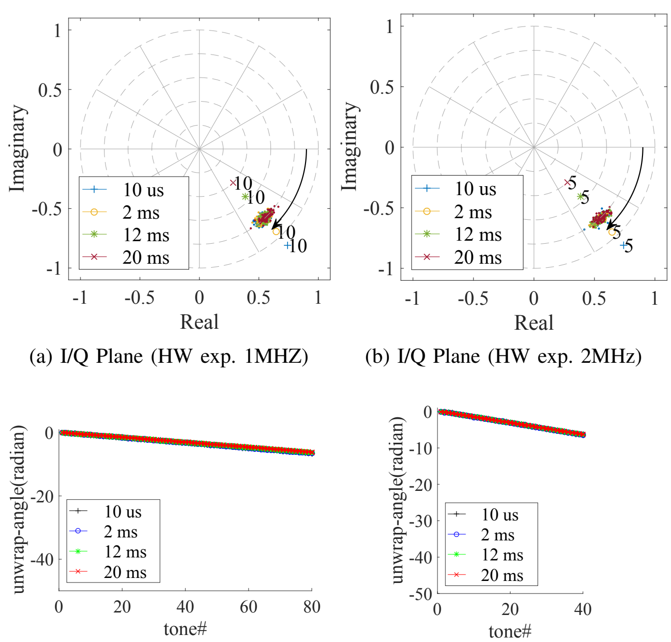
Two effects are noticeable when changing T_a; Firstly, the width of the measurement-cloud increases with T_a. With increasing T_a, both the LO initiator and reflector accumulate more phase-noise, leading to a less accurate phase measurement. Secondly, the center-of-mass of each cloud is rotating through the IQ-plane with increasing T_a. For example, tone#10 rotates its phase by -46.3°, -73.3°, -209.1° and -315.9° degrees for four T_as, respectively. Due to this rotation, the bias on the range-estimate will increase with T_a. This is in line with our model (13), which predicts a linear relationship between T_a and the bias of the ranging error with a ratio of $\alpha_f + \alpha_o$. Note that the expected phase for the 10-th tone for different T_a can



(a) IQ Plane (HW exp. 1MHz) (b) IQ Plane (HW exp. 2MHz)

(c) HW experiment (1MHz) (d) HW experiment (2MHz)

Fig. 5. IQ Planes (tone # 10 and 5) and phase-frequency graphs for HW experiment with different T_a values. (a) and (c) have freq. step of 1MHz, (b) and (d) have freq. step of 2MHz (@2400-2480 MHz). (Range is proportional to phase-frequency slope $\Delta\varphi/\Delta f$)



(a) I/Q Plane (HW exp. 1MHZ) (b) I/Q Plane (HW exp. 2MHz)

(c) HW experiment (1MHz) (d) HW experiment (2MHz)

Fig. 6. IQ Planes (tone # 10 and 5) and phase-frequency graphs for HW experiment with different T_r values. (a) and (c) have freq. step of 1MHz. (b) and (d) have freq. step of 2MHz (@ 2400-2480 MHz). (Range is proportional to phase-frequency slope $\Delta\varphi/\Delta f$)

be found outside the unit-circle as individual, labelled dots. In addition, Figure 5(c) depicts the unwrapped angle for those four T_a values. We observe different slopes for $10\mu s$, $2ms$, $12ms$ and $20ms$ delays and as we increase the T_a value, the slope will increase. The slope (i.e., $\Delta\varphi/\Delta f$) is proportional to range, where $r = \frac{c_0}{4\pi}.\frac{\Delta\varphi}{\Delta f}$.

In Figure 5(c), after measuring the slope of lines, the estimated range will be $1.97m$, $3.07m$, $8.67m$ and $13.12m$ for those four T_a values. Figures 5(b) and (d) show similar results regarding the frequency step of 2MHz and 40 tones (three advertising channels and 37 data channels), where only tone #5 is shown, which results the same amount of frequency change as tone #10 did with the 1 MHz frequency step. For frequency step of 2 MHz, we observe similar behavior as in the frequency step of 1MHz.

B. *Inter-delay (T_r) increase and the effect of crystal offset on ranging error*

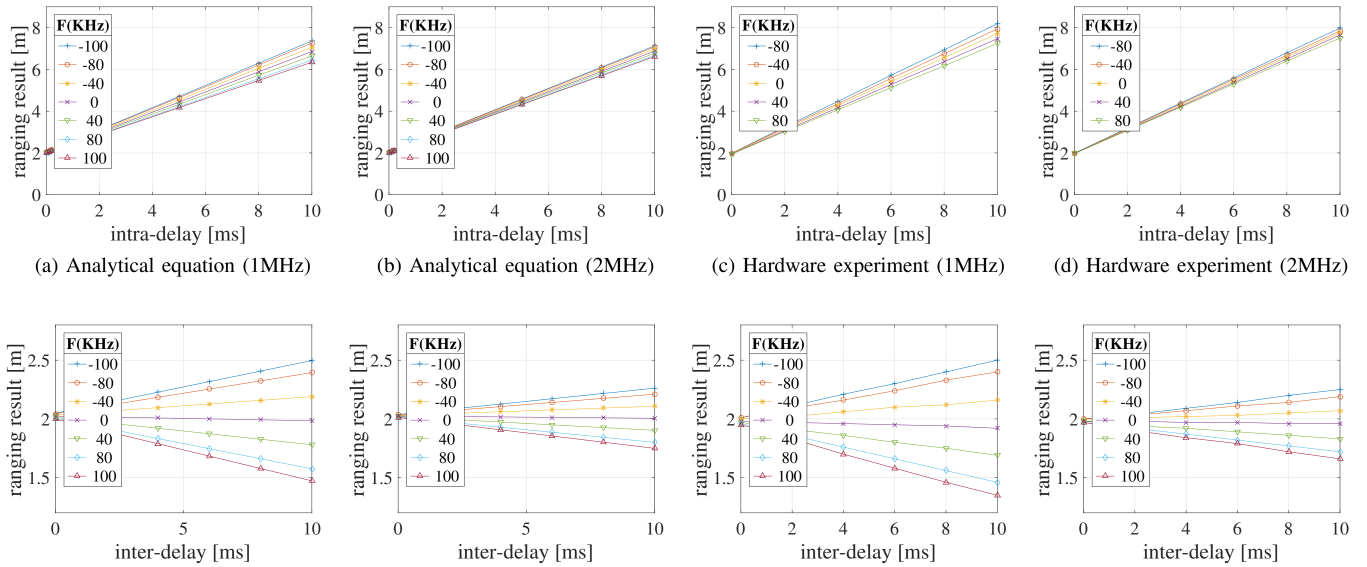
In addition, we studied the effects of increased inter-delays (T_r) on ranging results. Figure 6(a) depicts the I/Q planes for the collected phase measurement values for 80 tones with a frequency step of 1MHz. Whereas an increase in T_a expands the size of the I/Q cloud, the size of the I/Q cloud does not change when the T_r value is increased from $10\mu s$ to $20ms$. When increasing T_r, the phase measurement is not prone to more phase-noise. Consequently, the ranging accuracy will not be affected. Moreover, the I/Q cloud does not rotate and the slope of the unwrap angle remains the same. This confirms our findings in (13); an increase in T_r affects the ranging error by α_f. This increase is negligible, in contrast to the effect of T_a, which causes a large ranging error by $\alpha_f + \alpha_o$. In our experiment, the F is almost 0 kHz, T_a is very small ($10\mu s$) and crystal-offsets are 7.25 and 4 ppm, which results in α_f being at least a factor 248 smaller than α_0. In other words, the ranging error is almost independent of T_r and thus remains unaffected.

In addition, Figure 6(c) depicts the unwrap angle for those four inter-delay values. We observe similar slopes for $10\mu s$, $2ms$, $12ms$ and $20ms$ delays. Upon increasing the T_r, the slope will not change. Figures 6(b) and (d) show similar outcomes for the frequency step of 2MHz and 40 tones, where only tone #5 is shown to assure the same amount of frequency change as tone #10 does for the 1 MHz frequency step. We observe similar patterns in the frequency step of 1MHz.

C. *Intra- and inter-delay increase and the effect of software-controlled frequency offset (F) in terms of ranging error*

Figure 7(a-d) depicts the outcomes of the analytical equation and experiments used to evaluate the effects of programmed frequency offset (F) and T_a on the ranging result. In addition, Figure 7(e-h) shows the outcomes of the analytical equation and experiments used to evaluate the impact of programmed frequency offset (F) and T_r on ranging results. In Figure 7, the ranging result includes the effective cable length[4] ($\approx 2m$)

[4]Due to the lower propagation speed of RF signals through SMA cables, the measured distance is longer than the actual cable-length.



(a) Analytical equation (1MHz) (b) Analytical equation (2MHz) (c) Hardware experiment (1MHz) (d) Hardware experiment (2MHz)

(e) Analytical equation (1MHz) (f) Analytical equation (2MHz) (g) Hardware experiment (1MHz) (h) Hardware experiment (2MHz)

Fig. 7. Analytical equation (a,b,e, and f) and hardware experiment (c,d,g and h) results for evaluating the effect of varied intra-, inter-delays and software-controlled frequency offset (F) on ranging results.

and the ranging error. For the analytical equation, we use the equation introduced in Sec. IV-C to study the effects of delays. The results of the analytical equations reflect the outcomes of the experiment.

Figure 7(a-h) shows that varying T_a (phase-measurement offset) from $10\mu s$ to $10ms$ affects the ranging error up to 5 meters. This is much more than the effect of T_r on the ranging error, which amounts to approximately 1 meter. This behavior can be explained by (13) as well as the fact that α_0 is often significantly larger than α_f. This results in a range error that is more sensitive to T_a than to T_r.

In line with (13), the ranging error is equal to $(\alpha_f + \alpha_a)T_a + c \approx (\tilde{\alpha}_o + \tilde{\alpha}_o \frac{F}{\Delta_f})T_a + c$, for small T_r ($10\mu s$). The ranging error is directly proportional to software-controlled frequency offset (F) and inversely proportional to frequency step (Δ_f). This means that by increasing the F, the error-T_a slope will change accordingly, as shown in Figure 7(a-d). When increasing Δ_f from 1 MHz to 2 MHz, the ranging error becomes less sensitive to F.

Similarly, according to (13), the ranging error is proportional to F and inversely proportional to Δ_f. This means that by increasing the F, the error-T_r slope will change accordingly, as shown in Figure 7(e-h). By increasing Δ_f from 1 MHz to 2 MHz, the ranging errors are indeed halved for all values of F.

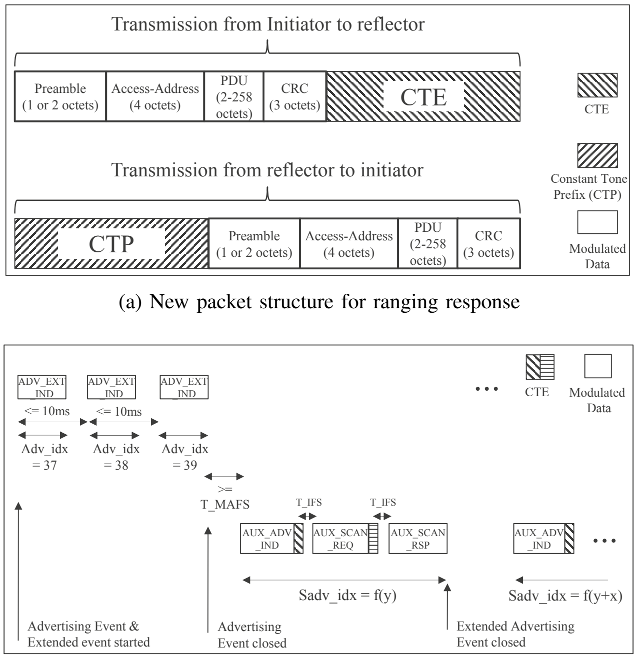
VI. Discussion

A. Front-bits for reflector responses

In both our connection-oriented and connectionless proposed solutions, we assumed that the Ranging_IND commands (connection-oriented), AUX_SYNC_IND and SCAN_REQ (connectionless) carry the CTE at the end of the packet. However, placing (phase)-modulated data in between two CTEs exchange, could cause a difference in phase between reception and transmission of the constant tone at the reflector, when not properly compensated. This problem can be avoided by defining a new packet structure for the ranging response, where the reflector sends the constant tone in front of the packet, named Constant Tone Prefix (CTP). Doing so has the additional benefit of further reducing the intra-delay T_a, which makes the ranging system less sensitive to phase-noise and crystal-offset. The model of Sec. IV-C also applies to the altered packet-structure; only the value of T_o is reduced. Figure 8(a) shows the newly proposed packet structure for ranging response. In the new packet structure, the preamble is no longer at the beginning the packet, which may effect the detection of the beginning of the reflector packet. However, if the timing between the request-response packets (i.e., T_IFS) is fixed, the initiator knows the beginning of the received packet.

B. Connectionless ranging by AUX_ADV_IND

In addition, the advertising packet (AUX_ADV_IND) on secondary advertising channels can also be used to perform tone-exchanges when there is a connectionless mode of operation. Similar to what we proposed for periodic advertising, in the secondary advertising mechanism, the scanning device (reflector) can transmit AUX_SCAN_REQ packets including CTE to perform phase-measurements upon reception of an AUX_ADV_IND packet including CTE from the advertiser (initiator). This process is shown in Figure 8(b). This procedure continues and in the next cycle a new secondary



(a) New packet structure for ranging response

(b) Connectionless ranging solution, based on Secondary Advertising Packets

Fig. 8. connectionless ranging with Secondary Advertising Packets

advertising channel is calculated to exchange the constant-tone between advertiser and scanning devices. This procedure is slower than tone-exchange in the periodic advertising mechanism introduced in Section IV-A. In the periodic advertising mechanism, T_r starts from 7.5 ms. In contrast, T_r in AUX_ADV_IND mode starts from 24.3ms ($T_{rx\_ADV\_EXT\_IND} * 3 + T_{rx\_MAFS} + T_{rx\_AUX\_ADV\_IND}$).

VII. Conclusion

In this article, we introduced a phase-based ranging solution compatible with the BLE standard that utilizes BLE channel hopping to exchange tones in the entire 2.4 GHz frequency band to mitigate the multi-path fading problem. Our analytical model describes the effects of BLE packet spacing (named intra-delay or T_a) and slow channel hopping (named inter-delay or T_r) on ranging accuracy. Our proposed solution is designed for both the connection-oriented and connectionless modes of operation in the BLE standard. Carried out hardware-experiments and analytical modeling exercises confirmed the validity and accuracy of our model. Our experiments showed that by increasing the time between phase-measurement at reflector and initiator (T_a) (unlike increasing T_r), more phase-noise will be accumulated in their LO that affects the ranging accuracy. In addition, we show that increasing T_a (unlike increasing T_r) is likely to have a bias on the ranging error.

We can conclude that according to (13) the ranging error is predictable. Most of the parameters in (12) and (11) are known to the system, including software-controlled frequency offset (F), time offset between phase-measurement at two sides (i.e., T_IFS and reflector response modulated packet size), the connection-interval of BLE, frequency step ($\Delta_f = 2\ MHz$), number of BLE channels ($K_f = 37$). The only unknowns are the crystal offsets of initiator and reflector, which cannot be measured without an absolute reference. Fortunately, (13) can be computed using approximations (14), (15), which require only the crystal offset difference ($\eta_r - \eta_i$). This can be obtained by using a frequency-offset measurement between the reflector and initiator, which can be mapped to a crystal-offset difference.

An additional way to reduce phase-noise and a smaller ranging error, is to ensure that the reflectors response is sent with a minimum modulated packet size (short PDU size) upfront the CTE, with the shortest possible intra-delay (T_IFS) and to maintain phase coherence between initiator request CTE and the reflector's CTE.

References

[1] S. He and S. . G. Chan, "Wi-fi fingerprint-based indoor positioning: Recent advances and comparisons," *IEEE Communications Surveys Tutorials*, vol. 18, no. 1, pp. 466–490, Firstquarter 2016.

[2] A. Ozer and E. John, "Improving the accuracy of bluetooth low energy indoor positioning system using kalman filtering," in *Computational Science and Computational Intelligence (CSCI), 2016 International Conference on*. IEEE, 2016, pp. 180–185.

[3] B. T. Fang, "Simple solutions for hyperbolic and related position fixes," *IEEE transactions on aerospace and electronic systems*, vol. 26, no. 5, pp. 748–753, 1990.

[4] H. Liu, H. Darabi, P. Banerjee, and J. Liu, "Survey of wireless indoor positioning techniques and systems," *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, vol. 37, no. 6, pp. 1067–1080, 2007.

[5] A. Bensky, *Wireless positioning technologies and applications*. Artech House, 2016.

[6] A. P. T. Kainulainen and J. J. M. Jantunen, "Additional data usable in apparatus positioning," Nov. 15 2016, uS Patent 9,494,673.

[7] J. J.-M. Wang, K.-C. Chou, C.-H. Yu, and H.-Y. Liu, "Direction finding antenna format," Feb. 18 2016, uS Patent App. 14/828,679.

[8] M. J. Honkanen, A. P. T. Kainulainen, J. J. Salokannel, and J. P. Reunamäki, "Facilitating positioning through bluetooth low energy wireless messaging," Feb. 24 2015, uS Patent 8,965,284.

[9] "Bluetooth core specification 5.1," Bluetooth Special Interest Group, Standard, Jan. 2019.

[10] Y. Liu, W. Qi, S. Zhang, and Z. Wang, "A phase-based ranging method for fast frequency hopping system," in *2014 Sixth International Conference on Wireless Communications and Signal Processing (WCSP)*, Oct 2014, pp. 1–5.

[11] E. Yuan, W. Qi, P. Liu, L. Wei, and L. Chen, "Ranging method for navigation based on high-speed frequency-hopping signal," *IEEE Access*, vol. 6, pp. 4308–4320, 2018.

[12] B. Xu, W. Qi, Y. Zhao, L. Wei, and C. Zhang, "Holographic radio interferometry for target tracking in dense multipath indoor environments," in *Wireless Communications and Signal Processing (WCSP), 2017 9th International Conference on*. IEEE, 2017, pp. 1–6.

[13] M. Maróti, P. Völgyesi, S. Dóra, B. Kusŷ, A. Nádas, Á. Lédeczi, G. Balogh, and K. Molnár, "Radio interferometric geolocation," in *Proceedings of the 3rd international conference on Embedded networked sensor systems*. ACM, 2005, pp. 1–12.

[14] B. Dil and P. J. Havinga, "Stochastic radio interferometric positioning in the 2.4 ghz range." in *SenSys*, 2011, pp. 108–120.

[15] J. Ryoo and S. R. Das, "Phase-based ranging of rfid tags with applications to shopping cart localization," in *Proceedings of the 18th ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems*. ACM, 2015, pp. 245–249.

[16] W. Kluge and E. Sachse, "Distance measurement between two nodes of a radio network," Jun. 23 2016, US Patent App. 15/057,076.

[17] M. Pelka, C. Bollmeyer, and H. Hellbrck, "Accurate radio distance estimation by phase measurements with multiple frequencies," in *2014 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, Oct 2014, pp. 142–151.